Automatic Border Crossing Detection and Navigation of Boat

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Abstract

The aim of the project is to help the fisherman for identify our Indian border in the sea area while fishing. By using the 8 bit microcontroller the specified tasks is used to identify the Lankan border. This project is help to know our border limit of the navy. Then to use this project we can avoid unnecessary shooting and unnecessary death. The fisher man can know their limits. The project consists of Transmitting base station, Receiver setup. The base station is placed at the Indo-Lankan border area. The receiver setup is placed at the boat. The transmitting base station transmits the RF signal The receiver setup is placed inside the boat receives the signal, when the boat is nearing the border line. When the receiver receives the signal, a Message is transmitted to the coastal guard by means of a warning alarm is given to the fisherman. By means of this instant action, the coastal guard can stop the fisherman in crossing the border.

Keywords: Microcontroller, Transmitting Base Station, RF Signal, Message, Receiver

I. INTRODUCTION

In this project we are using the wireless technology to send the message from the boat to the office with the position of the boat in the sea using the GPS (Global Positioning System) technology. We are using the computer to store the database of the boat and the details of the fisher men and also the details of the communication the objectives of the project is to determine the actual location of the boat and to turn the boat when it crosses a marked latitude and longitude. Also to communicate with the coastal guards about the location of the boat using GPS technology.

II. HARDWARE DESIGN

The system hardware of automatic border crossing detection consists of Power supply unit, RF Transmitter, RF Receiver Transmitting and Receiving Antenna, Microcontroller, Relay, Buzzer and DC Motor as shown in Figure.1 and Figure.2.

A. Transmitter Block Diagram:

This block consists of the transmitter section of automatic order crossing detection and navigation system.it acquires the actual position of the boat and transmits the signal by means of an antennae. This in turn is connected to a PIC16F877A microcontroller using RS232. The microcontroller stores the current location and is sent through the ZigBee transmitter module for further processing.

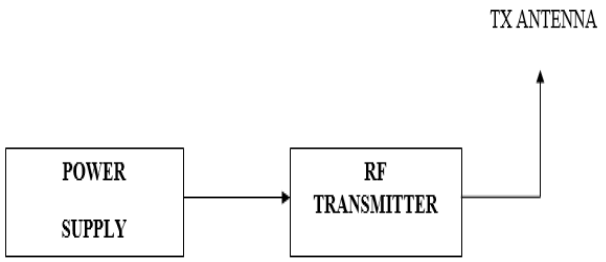


Fig. 1: Block Diagram of the Transmitter section

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B. Receiver Block Diagram:

The position of the boat is received by the Zigbee receiver module and is forwarded toPIC16F877A microcontroller. The microcontroller then sends the information to the GSM using MAX232 and further it is sent to the mobile through message if needed.

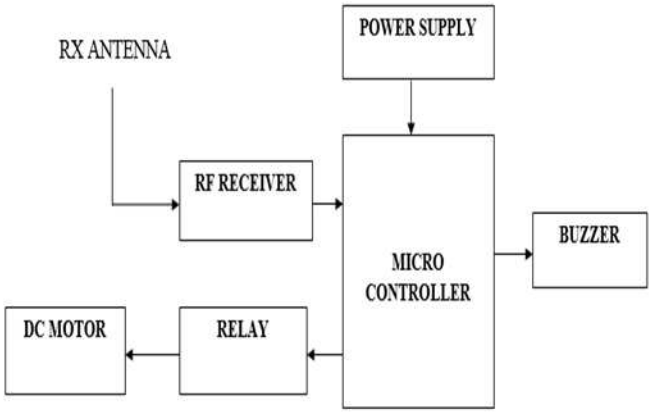


Fig. 2: Block Diagram of the Receiver section

C. Micro Controller:

A micro controller is a true computer on a chip. The design incorporates all the features found in a microprocessor such as CPU, ALU,PC, SP and registers. It also has some added features needed to make a complete computer ROM, RAM, parallel I/o, serial I/o, counters and clock circuit. The prime use of a micro controller is to control the operation of a machine using a fixed program that is stored in ROM and that does not change over the life time of the system. The areas if applications of micro controllers include control process, manufacturing process, medicine, instrumentation etc.

D. Pic Micro Controller:

PIC stands for peripheral interface controller as coined by microchip technology inc. USA. PIC is a very popular microcontroller worldwide. Microchip is the first manufacturer of 8 pin RISC MCU. Microchip is the world's second largest chip manufacturer. Focus on high performance cost-effective, field programmable embedded control solutions. Variety of end-user applications-specific standard products (ASSP) & application specific integrated circuits.

E. Power Supply Unit:

A power supply (sometimes known as a power supply unit or PSU) is a device or system that supplies electrical or other types of energy to an output load or group of loads. The term is mostly commonly applied to electrical energy supplies, less often to mechanical ones, and rarely to others. The transformer steps up or steps down the input line voltage and isolates the power supply from the power line. The rectifier section converts the alternating current input signal to a pulsating direct current. However, as you proceed in this chapter you will learn that pulsating dc is not desirable. For this reason a FILTER section is used to convert pulsating dc to a purer, more desirable form of dc voltage.

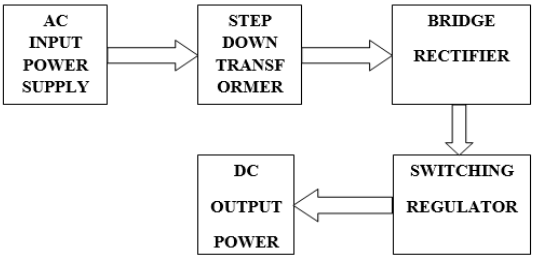


Fig. 3: Block Diagram of the Power Supply Unit

F. Step Down Transformer:

It is used to step down the main supply voltage by using step down transformer. It consists of primary and secondary coils. The output from the secondary is also AC wave form. So we have to convert AC voltage in to DC voltage by using rectifier unit. The transformer is based on two principles: first, that an electric current can produce a magnetic field (electromagnetism) and second, that the changing magnetic field within a coil of wire induces a voltage across the ends of the coil (electromagnetic induction).By changing the current in the primary coil one can change the strength of its magnetic field; since the secondary coil is wrapped around the same magnetic field, a voltage is induced across the secondary. A current passing through the primary coil creates a magnetic field. The primary and the secondary coil are wrapped around a core of very high magnetic permeability, such as iron.

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This ensures that most of the magnetic field lines produced by the primary current are within the iron and pass through the secondary coil as well as the primary coil.

G. Bridge Rectifier:

A rectifier is an electrical device that converts alternating current to direct current or at least to the current with only the positive value, a process is known as rectification .Rectifiers are used as components of power supplies and as detectors of radio signals. According to the conventional model of current flow still followed by most Engineers today, current is assumed to flow through electrical conductors from the positive to the negative pole. In actuality, free electrons in a conductor nearly always flow from negative to positive pole. In the vast majority of applications, however, the actual direction of the current flow is irrelevant. When the input connected to the left corner is negative and the input connected to the right corner is positive, current flows from the upper supply terminal to the right along the red (positive) path.

H. ZIGBEE Module:

The XBee®/XBee-PRO RF Modules interface to a host device through a logic-level asynchronous serial port. Through its serial port, the module can communicate with any logic and voltage compatible UART; or through a level translator to any serial device.Devices that have a UART interface can connect directly to the pins of the RF module. Each data byte consists of a start bit (low), 8 data bits (least significant bit first) and a stop bit (high). The following figure illustrates the serial bit pattern of data passing through the module. Serial communications depend on the two UARTs (the microcontroller's and the RF module's) to be configured with compatible settings (baud rate, parity, start bits, stop bits, data bits). The UART baud rate and parity settings on the XBee module can be configured with the BD and SB commands, respectively.

I. Buzzer:

A buzzer or beeper is a signaling device, usually electronic, typically used in automobiles, household appliances such as a microwave oven, or game shows. It most commonly consists of a number of switches or sensors connected to a control unit that determines if and which button was pushed or a preset time has lapsed, and usually illuminates a light on the appropriate button or control panel, and sounds a warning in the form of a continuous or intermittent buzzing or beeping sound. Initially this device was based on an electromechanical system which was identical to an electric bell without the metal gong (which makes the ringing noise). Often these units were anchored to a wall or ceiling and used the ceiling or wall as a sounding board. Another implementation with some AC-connected devices was to implement a circuit to make the AC current into a noise loud enough to drive a loudspeaker and hook this circuit up to a cheap 8-ohm speaker. Nowadays, it is more popular to use a ceramic-based piezoelectric sounder like a Sonalert which makes a high-pitched tone. Usually these were hooked up to "driver" circuits which varied the pitch of the sound or pulsed the sound on and off.The circuit is designed to control the buzzer. The buzzer ON and OFF is controlled by the pair of switching transistors (BC 547). The buzzer is connected in the Q2 transistor collector terminal. When high pulse signal is given to base of the Q1 transistors, the transistor is conducting and close the collector and emitter terminal so zero signals is given to base of the Q2 transistor. Hence Q2 transistor and buzzer is turned OFF state. When low pulse is given to base of transistor Q1 transistor, the transistor is turned OFF. Now 12v is given to base of Q2 transistor so the transistor is conducting and buzzer is energized and produces the sound signal.

J. Relays:

A relay is an electromagnetically operated switch. Many relays use an electromagnet to mechanically operate a switch but other operating principles are also used such as solid state relays. Relays are used where it is necessary to control a circuit by a low power signal or where several circuits must be controlled by one signal. A type of relay that can handle the high power required to directly control an electric motor or other loads is called a contactor. Solid state relays control power circuits with no moving circuits, instead using a semiconductor device to perform switching. Sometimes multiple operating coils are used to protect electrical circuits from overload or faults.

III. SOFTWARE TOOLS AND WORKING

A. MPLAB:

MPLAB IDE is an integrated development environment that provides development engineers with the flexibility to develop and debug firmware for various Microchip devices. MPLAB IDE is a Windows-based Integrated Development Environment for the Microchip Technology Incorporated PIC microcontroller (MCU). Create source code using the built-in editor. Assemble, compile and link source code using various language tools. An assembler, linker and librarian come with MPLAB IDE. C compilers are available from Microchip and other third party vendors. Debug the executable logic by watching program flow with a simulator, such as MPLAB SIM, or in real time with an emulator, such as MPLAB ICE. Third party emulators that work with MPLAB IDE are also available. Make timing measurements. View variables in Watch windows. Program firmware into devices with programmers such as PICSTART Plus or PRO MATE.

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B. MPLAB Simulator:

MPLAB SIM is a discrete-event simulator for the PIC microcontroller (MCU) families. It is integrated into MPLAB IDE integrated development environment. The MPLAB SIM debugging tool is designed to model operation of Microchip Technology's PIC microcontrollers to assist users in debugging software for these devices.

C. IC Programming:

The PRO MATE II is a Microchip microcontroller device programmer. Through interchangeable programming socket modules, PRO MATE II enables you to quickly and easily program the entire line of Microchip PICmicro microcontroller devices and many of the Microchip memory parts.PRO MATE II may be used with MPLAB IDE running under supported Windows OS's (see Read me for PRO MATE II.txt for support list), with the command-line controller PROCMD or as a stand-alone programmer

D. Compiler-High Tech C:

A program written in the high level language called C; which will be converted into PIC micro MCU machine code by a compiler. Machine code is suitable for use by a PIC micro MCU or Microchip development system product like MPLAB IDE.

E. Pic Start plus Programmer:

The PIC start plus development system from microchip technology provides the product development engineer with a highly flexible low cost microcontroller design tool set for all microchip PIC micro devices. The pic start plus development system includes PIC start plus development programmer and MPLAB IDE. The PIC start plus programmer gives the product developer ability to program user software in to any of the supported microcontrollers. The PIC start plus software running under MPLAB provides for full interactive control over the programmer.

IV. RESULTS AND DISCUSSION

Present paper is designed using PIC microcontroller. It is proposed to design an embedded system which is used for automotive security of the coastal border PIC16F874A/877A microcontroller is used for interfacing to various hardware peripherals. For doing so a PIC16F874A/877A microcontroller is interfaced serially to a GPS Transceiver. An EEPROM is used to store the mobile number. The hardware interfaced to microcontroller are LCD display, GPS Transceiver. The design uses RS-232 protocol for serial communication between the modems and the microcontroller. A serial driver IC is used for converting TTL voltage levels to RS-232 voltage levels. The simulation is carried out by using MPLAB design tool.

It is software for microprocessor simulation, schematic capture and printed circuit board (PCB) design. It is developed by Microchip technology. MPLAB is a software package for computer-aided design of electronic circuits. The transmitter section as shown in Figure.4 produces radio waves with the aid of antenna. The transmitter itself generates a RF alternating current that is applied to the antenna. The transmitter consists of a power supply with which the RF signals are generated.



Fig. 4: Hardware setup of the Transmitter section of Automatic Border Crossing Detection and Navigation of Boat

The package is a system of circuit simulation, based on models of electronic components made in PSpice. A distinctive feature of the package is the possibility of simulating the operation of programmable devices: microcontrollers, microprocessors, DSP

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and others. Additionally, the package Proteus Professional is a system design of printed circuit boards. MPLAB IDE is an integrated development environment that provides development engineers with the flexibility to develop and debug firmware for various Microchip devices. MPLAB IDE is a Windows-based Integrated Development Environment Incorporated PIC microcontroller (MCU). The receiver section as shown in Figure.5 receives the RF signal from the transmitter and sends to the microprocessor for further processing.



Fig. 5: Complete Hardware setup of the Automatic Border Crossing Detection and Navigation of Boat

V. CONCLUSION

In this project fishermen can easily identify the national sea borders and therefore preventing them from entering their area. Thus saving their lives and providing good relationship with the neighboring countries. Also, the piracy of ship can be easily brought under control. In future, EEPROM can be used to store the previous navigated positions upto 256 locations. Also the size of the kit can be reduced by using GPS and GSM on the same module of the GPS navigator. This increases the accuracy upto 3m.

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